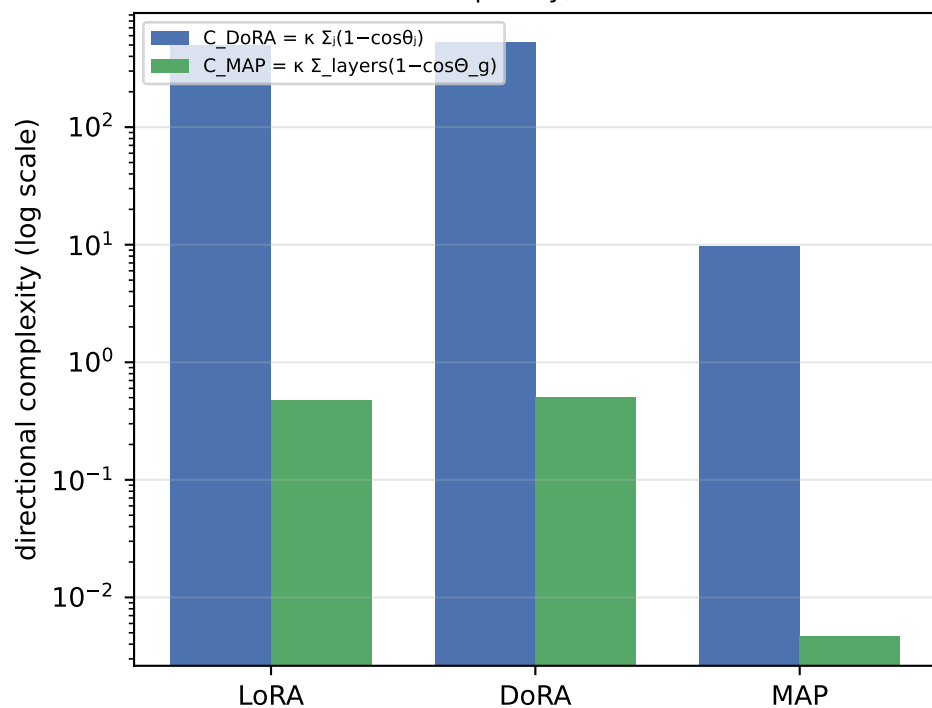


Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / SST2

Fine-tuning metrics and aggregate directional complexity

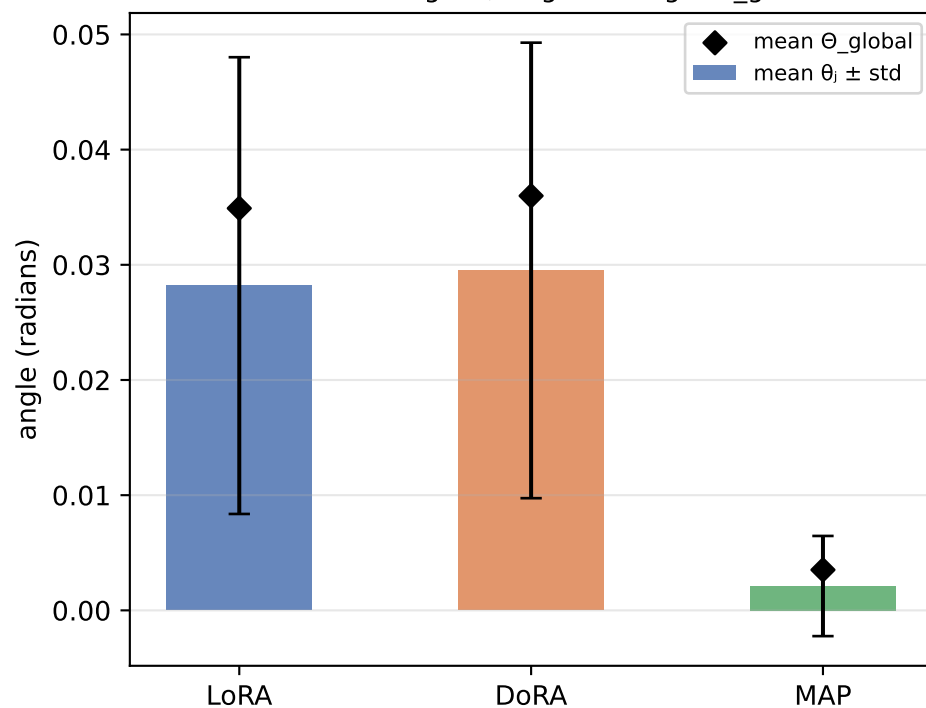
Method	Eval acc	Trainable params	C DoRA (Σ local)	C MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	94.50%	1,919,234	492.42	0.478	0.0282	0.0349	0.931
DoRA	94.61%	2,002,178	523.04	0.507	0.0295	0.0360	0.961
MAP	92.78%	1,919,378	9.58	0.005	0.0021	0.0035	0.007

Local (sum) vs global (single rotation) complexity, $\kappa=10$

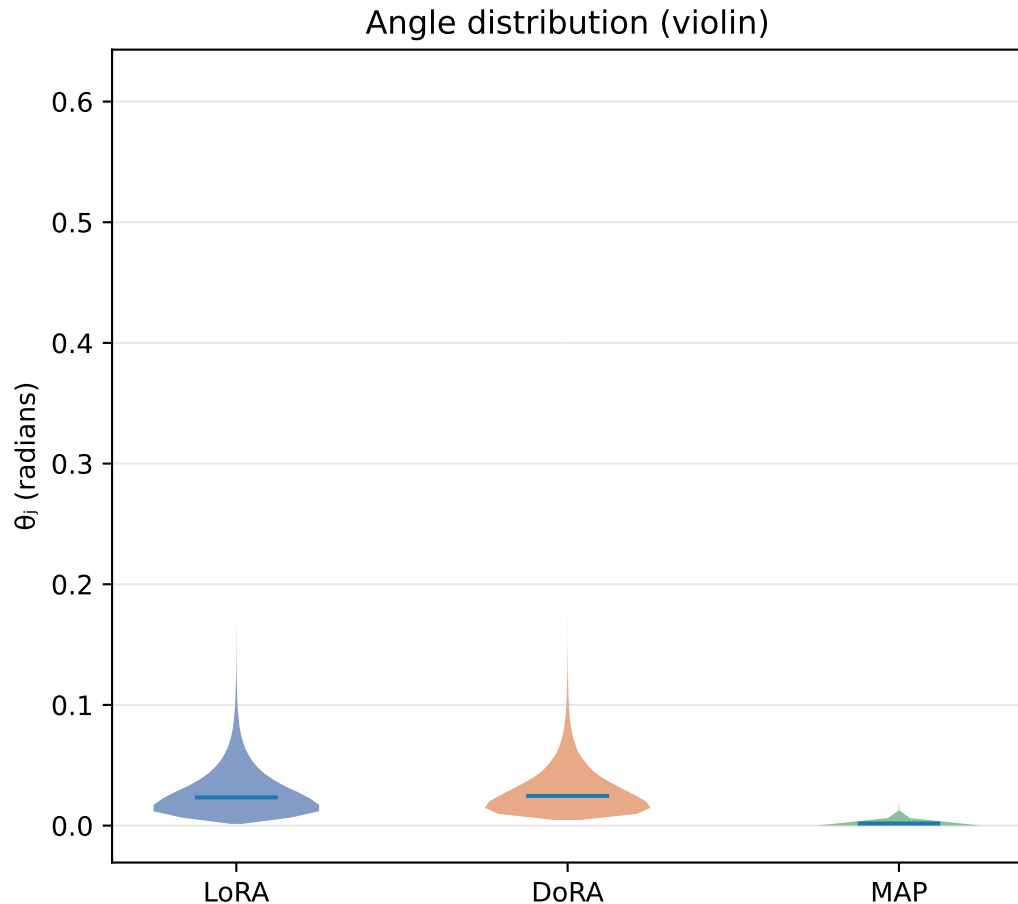
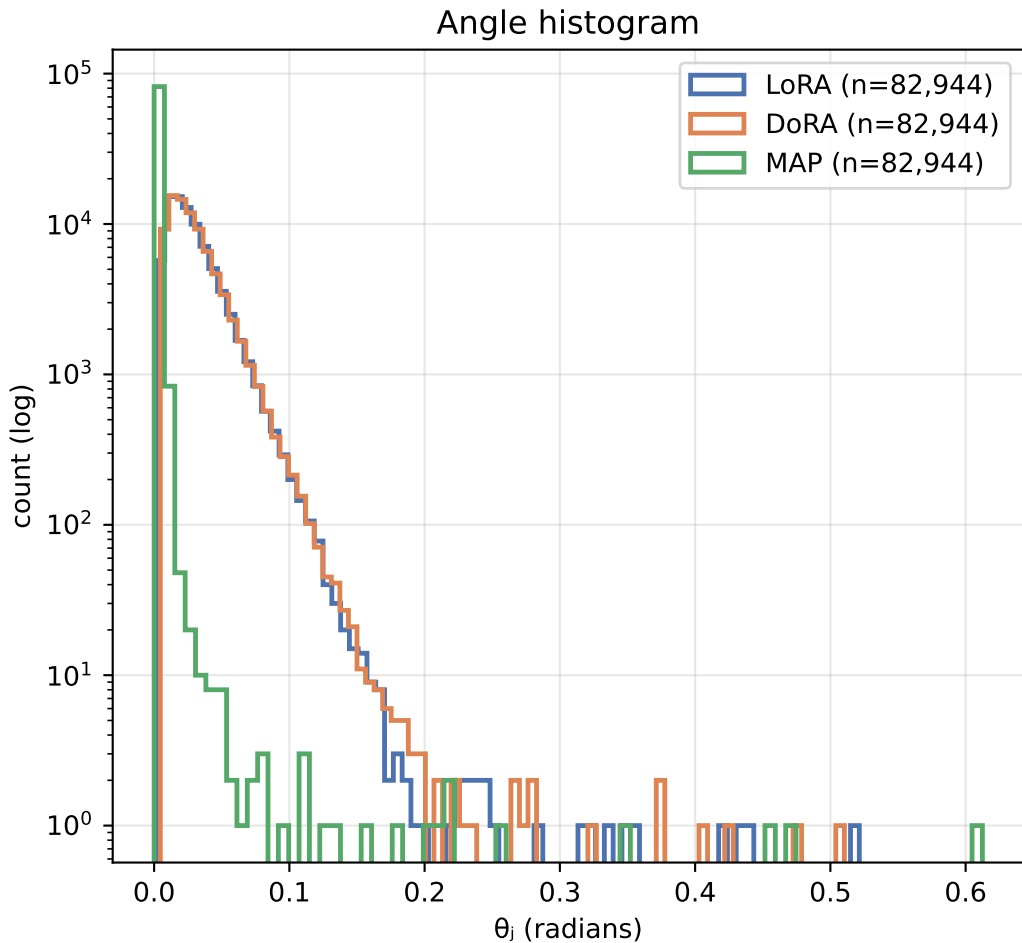


Θ_{global} is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1 - \cos \Theta_{\text{global}} = \text{mean}_j (1 - \cos \theta_j) \rightarrow \text{DoRA sums what MAP averages.}$

Per-column angle θ_j vs global angle Θ_{global}



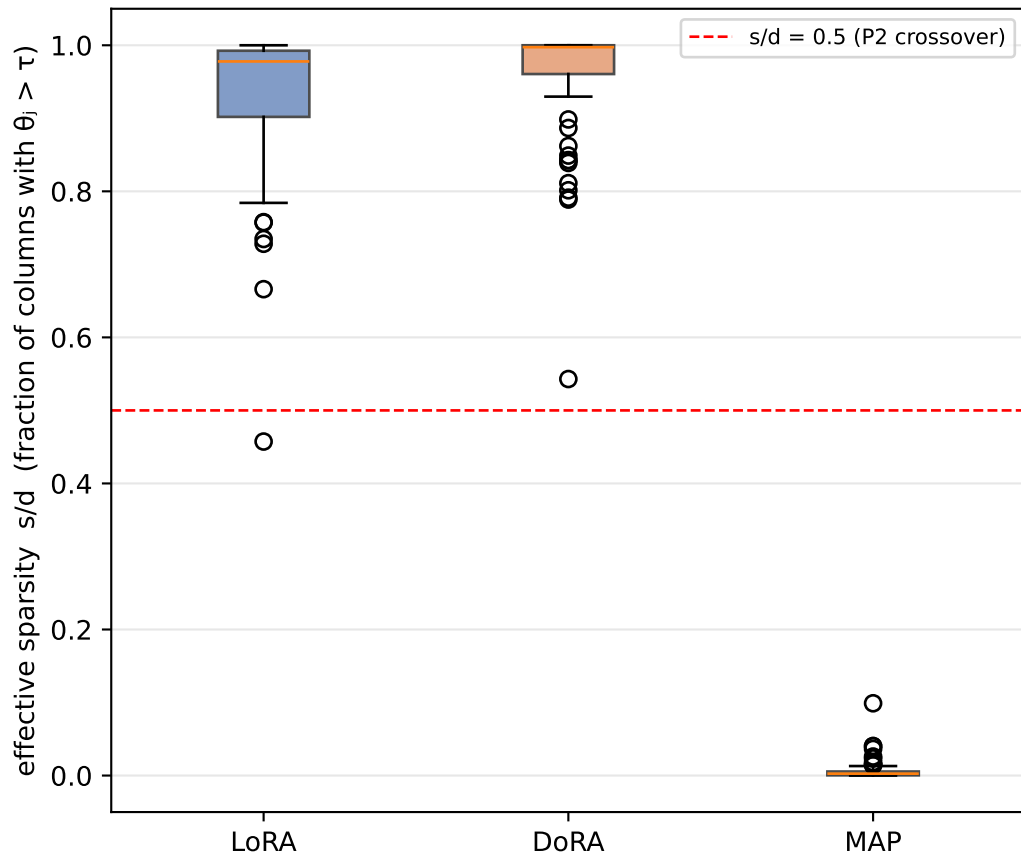
Distribution of per-column directional change θ_j (pooled over all target layers)



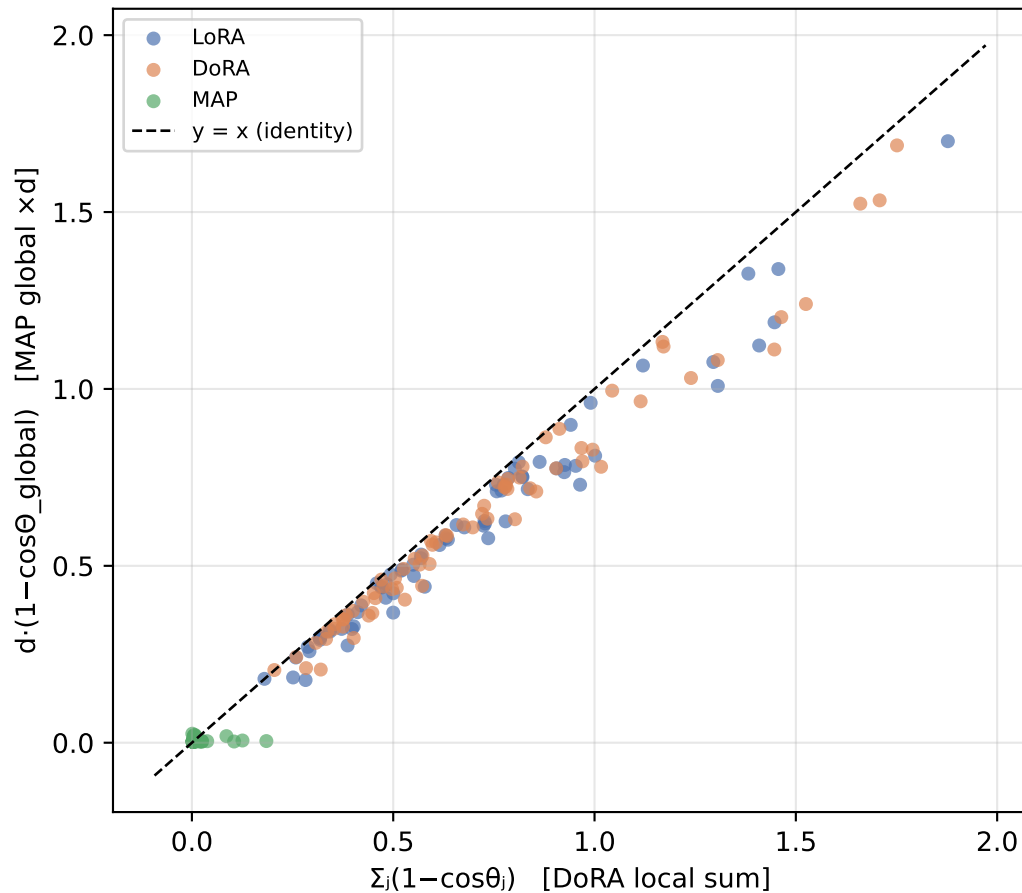
A long right tail = a few columns rotate a lot (sparse/localized update). A concentrated body near 0 = most columns barely move.

Sparsity of directional updates and the DoRA $\leftrightarrow$ MAP identity

Per-layer sparsity ($\tau=0.01$ rad)

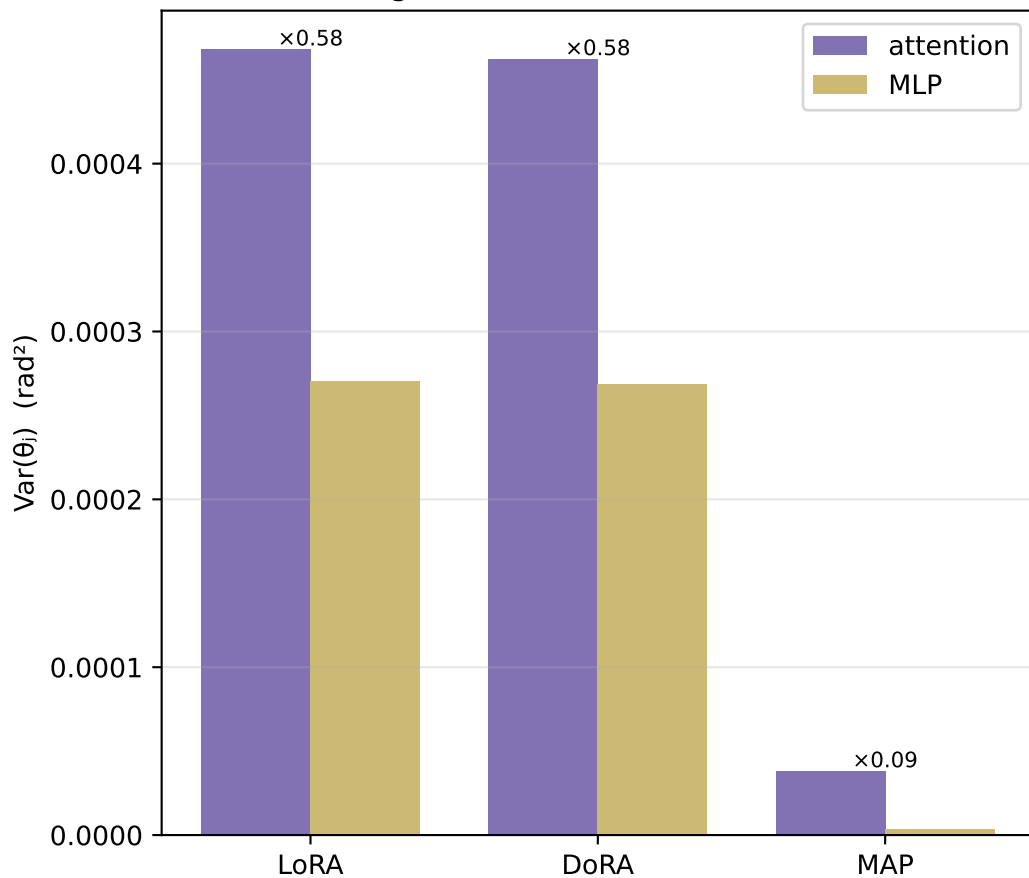


DoRA local sum vs MAP global $\times d$ (Thm 3.10 relation, approx: Θ_{global} uses raw — not unit — column flattening)



Where the rotation happens: attention vs MLP, and across depth

P3: angular variance, attention vs MLP



Mean per-column angle across depth

